SEQUENCE 4

Real numbers and intervals

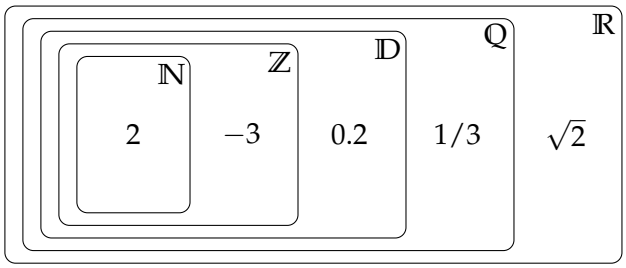
2–13 November 2026

4.1 Sets of numbers

Key point · Definitions.

- $\mathbb{N}$: natural numbers; $\mathbb{Z}$: integers.
- $\mathbb{D}$: decimal numbers, of the form $a/10^n$ with $a \in \mathbb{Z}$, $n \in \mathbb{N}$.
- $\mathbb{Q}$: rational numbers, of the form a/b with $a, b \in \mathbb{Z}$, $b \neq 0$.
- $\mathbb{R}$: real numbers, represented by the abscissae of the points of a graduated line. A real number that is not rational is **irrational**.

$$\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{D} \subsetneq \mathbb{Q} \subsetneq \mathbb{R}.$$



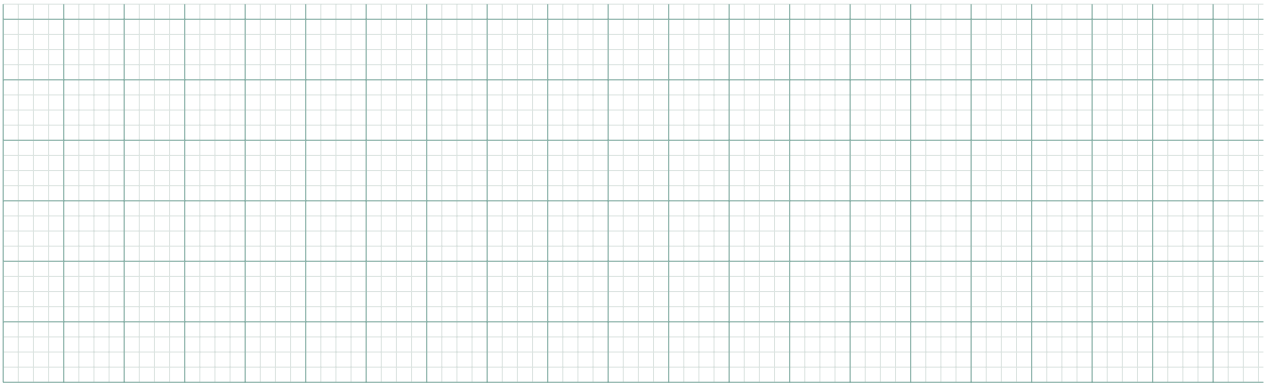
Example 1. Give the smallest of the five sets containing each of the numbers: -95; 7.29; $5/3$; $\sqrt{25}$; $\sqrt{2}$; π; 0.0002.

4.2 Two proofs by contradiction

Method · Proof by contradiction. To prove a statement, assume its negation, deduce a contradiction, then conclude that the statement is true.

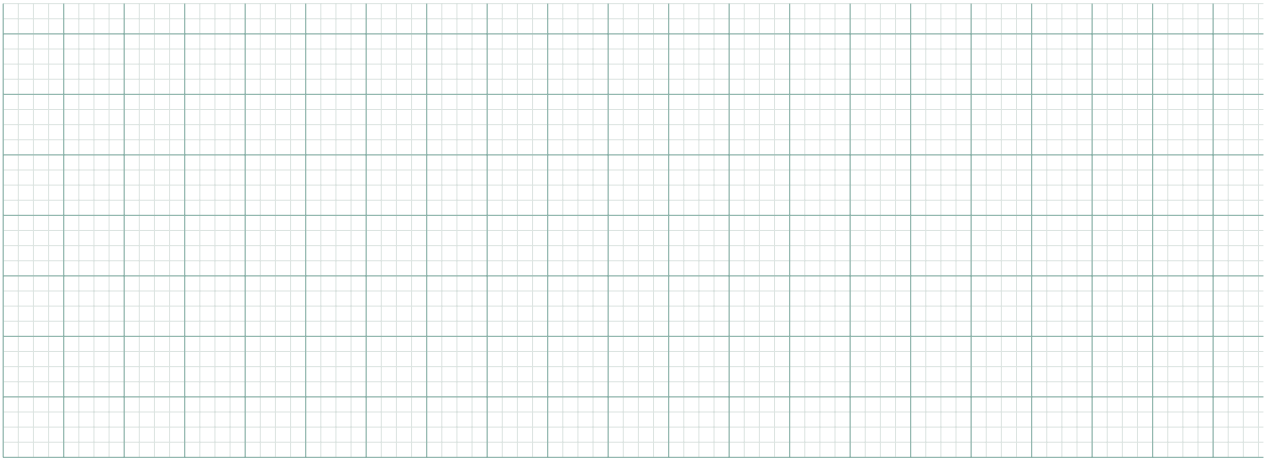
Key point · Property. $\frac{1}{3}$ is not a decimal number.

Proof to write.



Key point · Property. $\sqrt{2}$ is irrational.

Proof to write.



4.3 Intervals of the real line

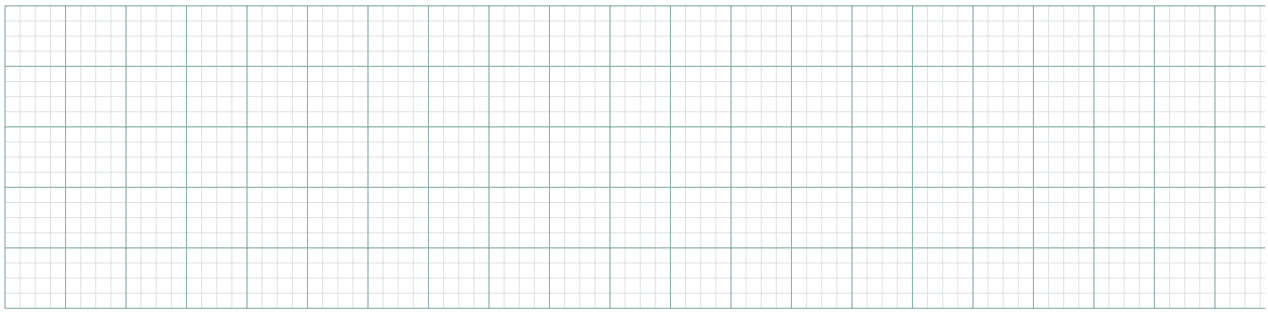
Key point · Notation. For two real numbers $a < b$, intervals describe the real numbers lying between two endpoints, with or without the endpoints. A square bracket includes the endpoint, a parenthesis excludes it. The symbols $-\infty, +\infty$ are not real numbers and always take a parenthesis.

Condition	Interval	Description
$a \leq x \leq b$	$[a, b]$	both endpoints included
$a < x \leq b$	$(a, b]$	a excluded, b included
$a \leq x < b$	$[a, b)$	a included, b excluded
$a < x < b$	(a, b)	both endpoints excluded
$x \geq a$	$[a, +\infty)$	greater than or equal to a
$x > a$	$(a, +\infty)$	strictly greater than a
$x \leq b$	$(-\infty, b]$	less than or equal to b
$x < b$	$(-\infty, b)$	strictly less than b

$$\mathbb{R} = (-\infty, +\infty), \quad \mathbb{R}^* = (-\infty, 0) \cup (0, +\infty).$$

Watch out. The strictly negative real numbers form $(-\infty, 0)$: 0 is excluded. $[a, a] = \{a\}$; $(a, a) = \varnothing$.

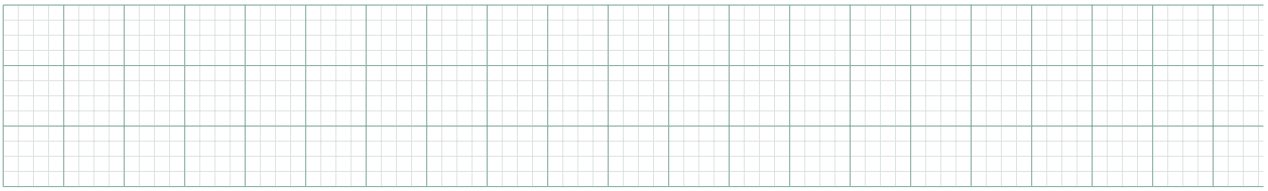
Example 2. Write as an interval then represent: $2 \leq x \leq 5$; $x > 4$; $-6 < x \leq 2$; $x < 0$.



4.4 Intersection and union of intervals

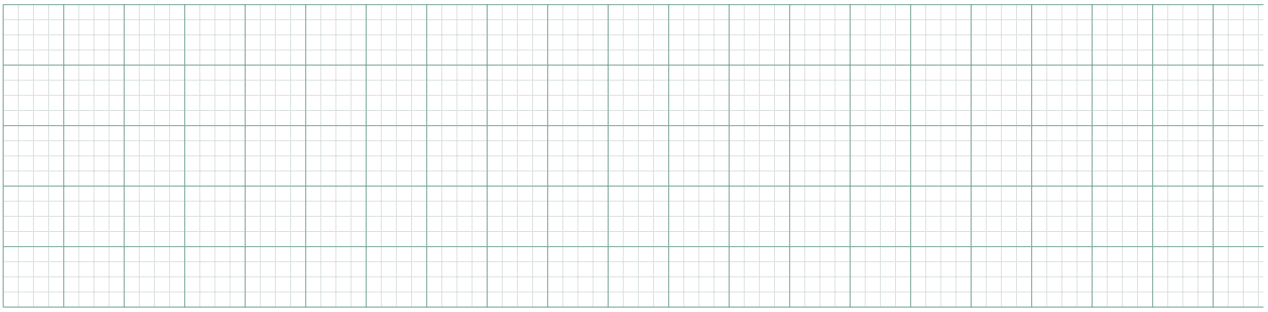
Key point · Definition. $I \cap J$ corresponds to the condition "belonging to I **and** to J"; $I \cup J$ corresponds to "belonging to I **or** to J". A union of intervals is not always an interval.

Example 3. With $I = [0, 12]$ and $J = [3, 20]$, determine $I \cap J$ and $I \cup J$.



Example 4. Complete, justifying with representations on a number line.

I	J	$I \cap J$	$I \cup J$
$(-4, 3]$	$[1, 5]$	.....	.....
$(-\infty, 2]$	$[-4, +\infty)$	.....	.....
$(-\infty, 7]$	$[7, +\infty)$	.....	.....
$(-\infty, 5]$	$[-3, +\infty)$	.....	.....
$(-2, 0)$	$(0, 3]$	.....	.....



Fluency · 5–10 min. Simplify $\overrightarrow{AB} + \overrightarrow{BC}$; solve $3x - 6 = 0$; calculate 10^{-2}.

4.5 Absolute value and distance

Key point · Definition and properties. $|x|$ is the distance from the real number x to 0; $|x| \geq 0$ and

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

The distance between a and b is $|b-a|$. For $a \in \mathbb{R}$ and $r \geq 0$:

$$|x-a| \leq r \iff x \in [a-r, a+r], \qquad |x-a| < r \iff x \in (a-r, a+r).$$

Example 5. Calculate $|3.2|$, $|-1.6|$, $|\sqrt{3}-2|$, then the distances between 2 and 5, 2 and -5, -2 and -5.

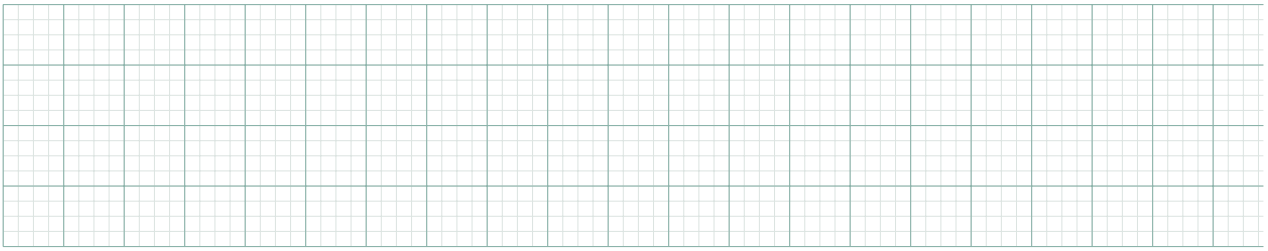
Example 6. Solve: $|x-7/3| = 14$; $|x-2| \leq 3$; $|x+3| < 5$.

Example 7. Write $x \in [4, 10]$ as an inequality with an absolute value.

4.6 Bounds and approximations

Key point · Vocabulary. To bound x is to give $a \leq x \leq b$; the width is $b - a$. An approximation from below is less than or equal to x; an approximation from above is greater than or equal to it. Rounding chooses the nearest value at the required precision. Significant figures are counted from the first nonzero digit; the displayed precision must suit the context.

Example 8. Bound $\sqrt{2}$ between two consecutive hundredths. Give its rounding to the nearest hundredth. Round 0.004567 to three significant figures, then 12345 to three significant figures in scientific notation.



```
def encadre_racine2(p):
    # p entier naturel ; calculs entiers exacts
    q = 10**p
    k = 0
    while (k+1)**2 <= 2*q*q:
        k = k+1
    return k/q, (k+1)/q
```

Example 9. Explain why this function bounds $\sqrt{2}$. What does it return for $p = 2$? Why does it terminate?

